An Event-Triggered Tube-MPC Framework for Aerial–Ground Tracking: Design and Computational Evaluation

Shlok Mehndiratta[1], Divyam Sood[1], and Prashant Gupta[1]

***Abstract*— We present an event-triggered tube-MPC framework for aerial–ground cooperative tracking, integrated end-to-end in ROS 2/Gazebo (software-in-the-loop), whose perception-driven trigger reduces online planner solves by $\approx 11\times$ at statistically equal tracking accuracy in our evaluation. A quadrotor UAV (abstracted as a velocity-controlled point mass) is the sole exteroceptive sensor for a unicycle UGV observed through a downward camera and an ArUco marker. The design couples perception and control through four implemented components: (i) a belief-space nonlinear MPC for the UGV that propagates the EKF covariance through its horizon and penalizes predicted covariance trace and field-of-view (FoV) exit; (ii) an ancillary tube regulator for the UAV that uses an ellipsoidal invariant-set bound whose radius scales with the EKF covariance; (iii) a multi-condition event trigger that replans on tube exit, covariance spikes, anticipated path curvature, or temporal drift; and (iv) a view-quality-dependent EKF measurement model. A minimal robust positively invariant (mRPI) set is derived offline as a certified robust bound but is *not* wired into the online loop, which uses the ellipsoidal bound of (ii). Our validated quantitative result is *computational*: across 8 paired runs the event trigger reduces online planner solves by $\approx 11\times$ ($331 \pm 3 \rightarrow 29 \pm 1$ per run) at statistically indistinguishable tracking error. The trigger is observability-aware *by design*; we claim only this compute payoff, and we document (Sec. VI) why an isolated observability benefit of the trigger is not demonstrable on the present platform. Throughout we mark which properties are proven offline, implemented, shown in simulation, or future work.**

I. INTRODUCTION

The deployment of Unmanned Ground Vehicles (UGVs) in unstructured and GPS-denied environments is heavily constrained by their limited perception horizons. Ground-level sensors such as LiDAR and stereo cameras suffer from occlusion, narrow fields of view (FoV), and an inability to anticipate upcoming path curvature. To mitigate this, aerial–ground cooperative systems employ an Unmanned Aerial Vehicle (UAV) as an elevated "eye in the sky" for a perception-limited UGV.

Most cooperative-tracking frameworks treat the UAV as a *passive* observer: it follows a predefined trajectory or simply minimizes distance to the UGV, treating the UGV's state estimation as a downstream byproduct [3]. This decoupling is fragile: when the UGV maneuvers aggressively or is occluded, the UAV can lose the target, the estimation covariance grows, and tracking degrades. Active-perception planning, where the sensor trajectory is optimized to maximize information gain [4], addresses part of this but typically relies on continuous, computationally expensive replanning and rarely integrates robust tracking guarantees for the aerial vehicle.

This paper studies a *bidirectional* observability-aware architecture in which *both* vehicles adapt to the ground vehicle's estimation uncertainty. Its contributions, as realized in the accompanying open implementation, are:

- **An observability-driven event-triggered tube-MPC design** integrating a covariance/FoV-penalizing belief-space UGV controller [5], [6], a covariance-scaled ellipsoidal tube regulator [1] (with an offline-certified mRPI bound [2]), and a multi-condition perception trigger [7] on tube exit, covariance spikes, anticipated curvature, or temporal drift. The covariance-scaling and FoV terms are design properties; we do *not* claim they are shown to improve observability (Sec. VI). [IMPLEMENTED]
- **A full SITL integration** in ROS 2/Gazebo — vision/ArUco pipeline, view-quality-coupled EKF, and coupled UGV/UAV controllers — demonstrated to run the design end-to-end; integration alone does not evidence efficacy of the coupling. [IMPLEMENTED]
- **A validated computational result**: the perception-driven trigger cuts online planner solves by $\approx 11\times$ at equal tracking accuracy (reproducible 6 m regime, 8 paired runs). This solve-count reduction is the validated payoff; the observability benefit of the covariance scaling is *not* isolated (Sec. VI). [SIMULATION]

The trigger's *design* is observability-aware; its *demonstrated* payoff is computational. We keep these separate throughout, and Sec. VI states precisely why the isolated observability benefit is not testable at the present scope.

We deliberately separate the idealized robust-MPC theory (offline mRPI, tightened constraints, terminal-set recursive feasibility) from what the code enforces online, and we validate the computational claim in simulation. The remainder of the paper is organized as follows. Section II reviews related work; Section III states the problem; Section IV gives the vehicle and perception models; Section V develops the coupled event-triggered tube-MPC; Section VI states the theoretical properties and the exact implementation status; and Section VII reports the simulation study.

*This work was carried out as part of the LCS-334 Linear Control Systems project.

[1]The authors are with the Indian Institute of Science Education and Research (IISER) Bhopal, India. {shlokm23, divyam23, prashant23}@iiserb.ac.in

II. RELATED WORK

Robust and tube MPC. Tube MPC [1] decomposes control into a nominal plan and an ancillary law that confines the true state to a robust positively invariant (RPI) tube; Raković et al. [2] give the standard outer approximation of the minimal RPI set by a truncated Minkowski series. These guarantees assume a fixed, bounded disturbance set and constraint tightening computed offline. Time-varying (homothetic/parameterized) tubes generalize the fixed tube but require the scaling to enter the optimization with its own feasibility argument; our online adaptivity (Section V) is a lighter-weight surrogate and we mark the homothetic treatment as future work.

Perception-aware and active planning. PAMPC [4] optimizes quadrotor trajectories for perception objectives. Belief-space planning [5], [6] augments the planning state with the estimator covariance. Our belief-space UGV controller is in this family. The *bidirectional* coupling — the *target* shaping its trajectory to stay observable while the *tracker* adapts its robustness to perception quality — is the design novelty of the architecture. What we *evidence* in this paper is narrower and concrete: the integrated SITL realization of that architecture and the perception-driven trigger's computational result (Sec. VII). Isolating the coupling's observability benefit is future work requiring a controllable-lag, trajectory-reproducible regime (Sec. VI).

Event-triggered MPC. Event-triggered schemes [7] resolve only when a state-dependent condition fires. We instantiate a perception-driven trigger with a critical tube-exit condition plus anticipatory conditions.

III. PROBLEM FORMULATION

A UAV tracks a UGV in a global frame $\mathcal{F}_W$. The UAV carries the only exteroceptive sensor: a downward monocular camera that detects an ArUco marker on the UGV, yielding a noisy pose measurement whose quality depends on the imaging geometry. An EKF fuses these detections with UGV wheel odometry to produce a belief (μ, Σ) over the UGV state. The objective is to (i) keep the UGV inside the camera footprint so that Σ stays bounded, (ii) track the UGV robustly under bounded disturbances, and (iii) do so while limiting the number of online optimizations. The UGV must track a ground lane while remaining observable.

IV. SYSTEM ARCHITECTURE AND MODELS

Fig. 1 shows the implemented system and its signal flow.

A. UGV dynamics

The UGV uses unicycle kinematics with first-order actuator lag. Let $\mathbf{x}_g = [x_g, y_g, \theta, v, \omega]^T$ be the state, (x_g, y_g) the position, θ the heading, (v, ω) the linear/angular velocities, and $\mathbf{u}_g = [v_c, \omega_c]^T$ the commanded velocities. The discrete-time dynamics are

$$\mathbf{x}_{g,k+1} = \mathbf{f}_g(\mathbf{x}_{g,k}, \mathbf{u}_{g,k}) + \mathbf{w}_g, \tag{1}$$

where the actuator lag is governed by time constants τ_v, τ_ω and $\mathbf{w}_g$ is process noise. [IMPLEMENTED]

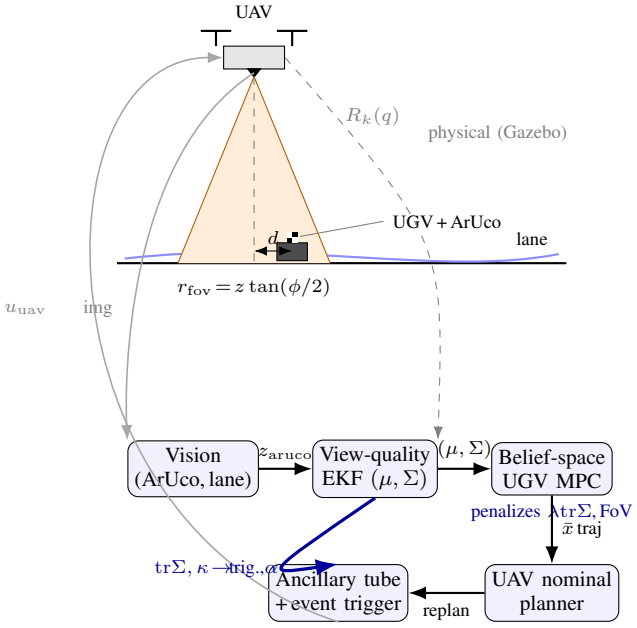


Fig. 1. Implemented system and signal flow. A downward-camera UAV (footprint radius $r_{\text{fov}} = z\tan(\phi/2)$) observes an ArUco-tagged UGV at horizontal offset d. **Black:** forward data flow (vision→EKF→belief-space UGV MPC→UAV nominal planner→ancillary tube/event trigger→UAV); **blue:** the perception→control couplings actually used — the UGV cost penalizes $\lambda\,\text{tr}\Sigma$ and FoV exit, and the EKF covariance/curvature drive the event trigger and tube radius α; **dashed grey:** the modeled dependence of measurement noise $R_k(q)$ on UAV geometry. All arrows are implemented ROS 2 signals; the figure asserts no performance claim.

B. UAV model

Although the platform is a quadrotor, the planner and the ancillary tracker treat it as a **velocity-controlled point mass** (single integrator); an inner-loop attitude/thrust controller, provided by the Gazebo velocity-control plugin, is assumed to realize commanded velocities. With position $\mathbf{p} = [p_x, p_y, p_z]^T$ and velocity command $\mathbf{u} = [v_x, v_y, v_z]^T$,

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \Delta t\,\mathbf{u}_k + \mathbf{w}_k, \qquad A_d = I_3,\ B_d = \Delta t\, I_3, \tag{2}$$

with bounded disturbance $\mathbf{w}_k \in \mathbb{W}$. Vertical velocity is a genuine decision variable, so altitude is planned rather than fixed. A second-order double-integrator (acceleration input) is the more faithful rigid-body abstraction; we use (2) because it is exactly what the implementation optimizes. [IMPLEMENTED]

C. View-quality-dependent perception

The camera's vantage point affects estimation accuracy. We use a differentiable view-quality metric $q_k \in (0, 1]$ that degrades as the UGV nears the FoV edge and as the altitude departs from a precision-optimal value z_{opt} (too low loses footprint; too high loses pixel resolution). With horizontal UAV–UGV distance $d = \|\mathbf{p}^{xy} - \mathbf{p}_g\|$ and footprint radius $r_{\text{fov}}(z) = z\tan(\phi/2)$,

$$q_k = \underbrace{\frac{1}{1 + e^{\beta(d - r_{\text{fov}})}}}_{\text{FoV gate}} \cdot \underbrace{e^{-d^2/(2\sigma_{xy}^2)}}_{\text{centering}} \cdot \underbrace{e^{-(z - z_{\text{opt}})^2/(2\sigma_z^2)}}_{\text{altitude precision}}, \tag{3}$$

with $\sigma_{xy} = r_{\text{fov}}/2$. The altitude factor makes q_k *non-monotone* in z (it does not tend to 1 as $z \to \infty$), removing the incentive to climb without bound. [IMPLEMENTED]

The measurement covariance grows as view quality drops. The implementation scales the baseline ArUco covariance R_{full} inversely with q_k,

$$R_k(q_k) = \frac{R_{\text{full}}}{q_k + \epsilon}, \qquad \epsilon = 10^{-3}, \tag{4}$$

so $R_k \to R_{\text{full}}$ for an excellent view and $R_k \to \infty$ (the update ignores the marker) as $q_k \to 0$. This couples UAV geometry directly to EKF performance. [IMPLEMENTED]

V. Coupled Event-Triggered Tube MPC

The estimation uncertainty Σ_k enters the control formulations of both vehicles.

A. Information-aware UGV control (belief-space NMPC)

The UGV controller augments the planning state with the covariance, $\xi_k = (\mu_k, \Sigma_k)$, and *propagates both through the horizon*. At each predicted step it forms the no-observation prediction $\Sigma_{\text{pred}} = F\Sigma F^T + Q$ and a post-measurement covariance Σ_{obs} (Joseph update with R_k from (4)), and blends them by a smooth visibility gate $\gamma = 1/(1 + e^{\beta(d - r_{\text{fov}})})$:

$$\Sigma_k = (1 - \gamma)\,\Sigma_{\text{pred}} + \gamma\,\Sigma_{\text{obs}}. \tag{5}$$

The cost minimizes tracking error while penalizing predicted uncertainty and FoV exit,

$$J_g = \sum_{k=0}^{N_g - 1} \Big(\|\mathbf{e}_{g,k}\|_Q^2 + \|\mathbf{u}_{g,k}\|_R^2 + \lambda\ \text{tr}(\Sigma_k) + w_{\text{fov}}\, c_{\text{fov}}(\mathbf{x}_{g,k}) \Big), \tag{6}$$

where the FoV term is a one-sided quadratic hinge $c_{\text{fov}} = \big(\max(0,\, d_k - r_{\text{fov}})\big)^2$ (a soft constraint, not an asymptotic barrier). The covariance rollout (5) makes (6) a nonconvex NLP, solved with SLSQP over $N_g = 5$. The two terms $\lambda\,\text{tr}(\Sigma_k)$ and $w_{\text{fov}} c_{\text{fov}}$ are what make the controller belief-coupled and are absent from the passive baseline. [IMPLEMENTED]

B. Aerial nominal planner

The UAV computes a nominal trajectory $\bar{\mathbf{p}}$ over $N_a = 10$ by optimizing goal tracking, an altitude regularizer toward z_{opt}, and a visibility term that tolerates lateral offset only when altitude is sufficient:

$$J_a = \sum_k w_{\ell,k} \Big[\|\bar{\mathbf{p}}_k - \mathbf{g}\|^2 + w_z (z_k - z_{\text{opt}})^2 + w_{\text{alt}} \frac{\|\bar{\mathbf{p}}_k^{xy} - \mathbf{g}^{xy}\|^2}{z_k^2 + 0.1} \Big] + w_u \|\bar{\mathbf{u}}_k\|^2, \tag{7}$$

where $\mathbf{g}$ is the UGV's predicted end-of-horizon position (read from the belief-MPC rollout, one direction of the coupling) and $w_{\ell,k}$ up-weights later steps. The weights are fixed (not re-scaled online); "climb into turns to widen the footprint" is an emergent optimum of (7), not a hand-coded rule. [IMPLEMENTED]

C. Offline robust invariant set (tube) derivation

The ancillary law is $\mathbf{u}_k = \bar{\mathbf{u}}_k - K\mathbf{e}_k$ with tracking error $\mathbf{e}_k = \mathbf{p}_k - \bar{\mathbf{p}}_k$; substituting into (2),

$$\mathbf{e}_{k+1} = \underbrace{(A_d - B_d K)}_{=:A_K} \mathbf{e}_k + \mathbf{w}_k, \qquad \mathbf{w}_k \in \mathbb{W}. \tag{8}$$

Unlike standard tube MPC, we account for measurement noise $\mathbf{v}_k$ entering through the feedback gain; the effective disturbance is $\mathbf{w}_{\text{eff}} = \mathbf{w}_a - B_d K \mathbf{v}_k$, and for the offline computation $\mathbf{v}_k$ is bounded to a box (a hard, undocumented-confidence truncation of the otherwise Gaussian noise; see Section VI). For A_K Schur and $\mathbb{W}$ compact, the minimal RPI set is $\Omega_\infty = \bigoplus_{i=0}^{\infty} A_K^i \mathbb{W}$, which we outer-approximate by truncating after $N = 100$ terms and scaling the support values by $1/(1 - \rho^{N+1})$, ρ the spectral radius of A_K, to conservatively bound the neglected tail [2]:

$$\Omega_N = \bigoplus_{i=0}^{100} A_K^i \mathbb{W}, \qquad \Omega \supseteq \Omega_\infty. \tag{9}$$

The state/input constraints are then tightened by the Pontryagin difference, $\bar{\mathbb{X}} = \mathbb{X} \ominus \Omega$, $\bar{\mathbb{U}} = \mathbb{U} \ominus K\Omega$. (9) and the tightening are computed offline and constitute a *certified robust bound for the idealized design*. [PROVEN-OFFLINE]

D. Online adaptive ellipsoidal tube

Rather than propagate the polytope Ω online, the ancillary controller uses an *ellipsoidal* invariant-set surrogate

$$\mathcal{E}(\alpha_k) = \{\mathbf{e} : \mathbf{e}^T P_0\, \mathbf{e} \le \alpha_k^2\}, \tag{10}$$

with a shape matrix P_0 (loaded from the offline design, else a Schur-consistent fallback) and a radius that adapts to the EKF horizontal covariance trace $\text{tr}\,\Sigma_{xy} = \Sigma_{xx} + \Sigma_{yy}$:

$$\alpha_k = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\,\text{clip}\Big(\tfrac{\text{tr}\,\Sigma_{xy} - \text{tr}_{\min}}{\text{tr}_{\max} - \text{tr}_{\min}}, 0, 1\Big), \tag{11}$$

($\alpha_{\min} = 0.5$, $\alpha_{\max} = 2$). The tube *breathes*: it tightens when the estimate is sharp and loosens when perception degrades. We stress that (11) scales the *invariant-set membership/trigger radius* used online; it is *not* accompanied by an online re-derivation of (9) nor by re-tightening $\bar{\mathbb{X}}, \bar{\mathbb{U}}$. A homothetic tube with a matching recursive-feasibility argument is future work (Section VI). [IMPLEMENTED]Accordingly, α_k is a heuristic monitor/trigger radius, not a certified invariant radius. The certified objects are the *offline* mRPI set Ω (9) and the A_K-Schur ISS bound of Sec. VI (which rests on the contraction A_K and bounded $\mathbb{W}$, not on α_k). The choice $\alpha_{\min} < 1$ tunes *trigger sensitivity* (how eagerly a replan fires), not invariance; letting the membership radius drop below the nominal invariant size therefore does not weaken those certified guarantees or contradict the offline mRPI bound.

E. Perception-driven event trigger

The ancillary node runs at 20 Hz but requests a nominal replan only when one of four prioritized conditions fires (implemented literally as follows).

- **(1) Tube exit (critical, no lockout):** fire iff $\mathbf{e}_k^T P_0 \mathbf{e}_k > \alpha_k^2$, i.e. the state left the adaptive ellipsoid (10).
- **(2) Covariance spike:** $\operatorname{tr}\Sigma_{xy}(t) > \tau_{\text{cov}}$ and a sudden ($>1.5\times$) rise over $\sim 0.5\,\text{s}$ (e.g. ArUco occlusion).
- **(3) Curvature anticipation:** discrete curvature of the UGV predicted path exceeds τ_κ, so the UAV can climb before a turn.
- **(4) Temporal drift:** elevated $\operatorname{tr}\Sigma_{xy}$ with time since last replan exceeding $T_{\min}$.

Conditions (2)–(4) carry a 1 s time-lockout (the chattering guard); condition (1) has none. Between triggers the ancillary law regulates the error via (8), and the previous nominal sequence is shifted forward.

We note that a purely fractional trigger on the *polytopic* tube, $\max(H_\Omega \mathbf{e}_k - \sigma h_\Omega) > 0$ with contracted set Ω_σ, is the natural formulation for the offline mRPI Ω and would furnish the tunable computation–accuracy parameter σ; the implementation instead uses the ellipsoidal test above, so we present the σ-polytopic trigger as future work rather than as an implemented result. [FUTURE WORK]

F. Nominal optimization actually solved

At each trigger the planner solves (7) over the single integrator (2) with velocity bounds $v_x, v_y \in [-1.5, 1.5]$, $v_z \in [-1, 1]$ m/s, warm-started from the previous solution (SLSQP, maxiter $= 15$). Altitude limits $z \in [z_{\min}, z_{\max}]$ are shaped by the cost; a hard terminal set $\mathbb{X}_f$ and the DARE terminal cost of the idealized formulation below are *not* enforced online. [IMPLEMENTED]

VI. Theoretical Properties and Implementation Status

We separate what is proven, what is inherited under assumptions, and what is only empirical.

Idealized recursive feasibility. For the error system (8) with Ω a true RPI set and tightened constraints $\bar{\mathbb{X}}, \bar{\mathbb{U}}$, a nominal MPC with a control-invariant terminal set $\mathbb{X}_f$ and DARE terminal cost is recursively feasible and the true state satisfies $\mathbf{x}_k \in \bar{\mathbb{X}} \oplus \Omega \subseteq \mathbb{X}$ [1], [2]. This holds for the *idealized* design; the code uses the ellipsoidal surrogate (10) and does not enforce tightening or a terminal set, so online recursive feasibility is *assumed, not verified*. [PROVEN-OFFLINE]

Bounded inter-trigger error (ISS). With A_K Schur and $\mathbf{w} \in \mathbb{W}$ bounded, (8) is input-to-state stable. For the implemented contraction $A_K = 0.6\, I_3$ this is immediate: $\|\mathbf{e}_{k+1}\| \le 0.6\|\mathbf{e}_k\| + \sup\|\mathbf{w}\|$, hence $\limsup_k \|\mathbf{e}_k\| \le \sup\|\mathbf{w}\|/0.4$. Thus, between triggers, the error stays bounded and the ellipsoid membership is a meaningful monitor. [PROVEN-OFFLINE]

Belief-MPC stability. Adding $\lambda \operatorname{tr}\Sigma_k$ and c_{fov} to the stage cost (6) makes it nonconvex and state-dependent; the standard terminal-cost machinery does not directly apply and we do not claim a formal stability proof for the UGV controller. [FUTURE WORK]

Stochastic/bounded mismatch. The mRPI machinery needs a bounded $\mathbb{W}$, whereas the EKF noise is Gaussian. The offline computation truncates $\mathbf{v}_k$ to a box without a stated confidence level, and the online Σ is never mapped to a hard bound. A probabilistic/stochastic-tube treatment is future work. [FUTURE WORK]

Table I summarizes the claim status.

TABLE I
CLAIM STATUS: PROVEN OFFLINE VS. IMPLEMENTED VS. FUTURE

Claim	Status
Belief NMPC propagates Σ, penalizes $\operatorname{tr}\Sigma$+FoV	implemented
View-quality-coupled R_k (4)	implemented
Non-monotone view quality in altitude (3)	implemented
Adaptive ellipsoidal tube radius (11)	implemented
Four-condition perception event trigger	implemented
Offline mRPI outer bound (9), tightening	proven (offline)
Bounded inter-trigger error / ISS	proven (offline)
σ-polytopic fractional trigger, Ω_σ	future work
Online mRPI / homothetic tube / re-tightening	future work
Terminal-set + DARE recursive feasibility online	future work
UGV belief-MPC formal stability proof	future work

Limitations and honest scope. The study is in simulation only; the UAV is a point-mass single integrator; the tube adaptivity is in the trigger/monitor radius, not a re-derived invariant set; the mRPI is offline and not wired into the online loop; and no online recursive-feasibility certificate is enforced. Most importantly, *the isolated observability benefit of the perception trigger is not demonstrable on this platform at the present scope*, for two documented reasons. (a) The UAV tracks the UGV's horizontal position tightly enough (mean lag $0.24\,\text{m}$ against ground truth) that the target stays near the optical center regardless of FoV size, so the observability margin is not stressed without changing control gains or vehicle speeds. (b) The regime that does shrink the margin — flying lower to reduce the footprint — is *not trajectory-reproducible*: with no seedable RNG and a wall-clock-paced, non-lockstep ROS 2/Gazebo loop, scheduling jitter in the closed perception–control–physics loop forks the UGV path by $2.8\,\text{m}$ on average ($4.9\,\text{m}$ max) between identical-configuration runs (target visibility 100% vs. 27%), so a fair perception-vs-baseline comparison cannot be run. Establishing the observability benefit is future work requiring (i) a controllable-lag regime (UAV speed/gain) that stresses the FoV margin, and (ii) a deterministic, lockstep simulator (`use_sim_time` with a fixed step) for seed-reproducible trajectories.

VII. Simulation Study

Setup. The system runs in Gazebo with ROS 2: a TurtleBot3-class UGV with an ArUco marker, an X500-class UAV with a downward camera, an OpenCV lane/marker pipeline, the 5-state EKF, the belief-space UGV MPC, and the UAV planner/ancillary pair. Per-step telemetry (pose, ground truth, $\operatorname{tr}\Sigma$, FoV distance, ArUco visibility, view quality, replan events, solve times) is logged by a common logger identical across configurations.

Determinism and experimental design. We first tested whether repeated runs under identical configuration produce

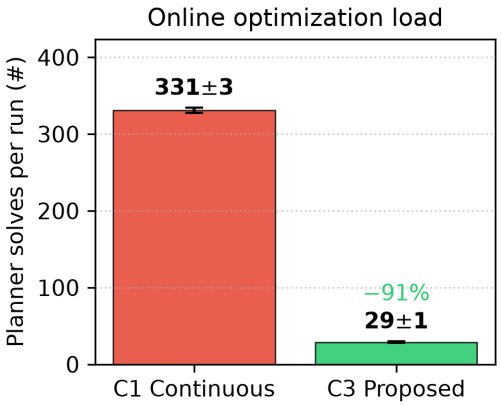


Fig. 2. Online planner solves per 40 s run (mean±std, 8 runs/method): continuous replanning (C1) vs. the perception-driven event trigger (C3), an $\approx 11\times$ (91%) reduction at statistically equal tracking error.

identical logs. They do *not*: two runs of the same configuration differed at the centimeter level in ground-truth pose and produced different sample counts, because the asynchronous ROS 2/Gazebo pipeline is not bit-reproducible and exposes no controllable seed. We therefore cannot pair noise realizations across methods; instead we take R repeats per (method, scenario) and report mean $\pm$ std (unpaired), across a nominal track and an injected-occlusion scenario.

Configurations. The UAV nominal planner consumes the UGV's predicted trajectory published by the belief-space MPC, so a fully passive UGV would leave the UAV planner without a reference; a decoupled passive-UGV baseline is therefore not a valid comparison in this implementation and is deferred to future work. We compare two functional configurations that differ only in the UAV replanning trigger, on identical scenarios and logging: **C1**, continuous replanning; and **C3**, the proposed perception-driven event-triggered replanning. An occlusion window is injected mid-mission via the ArUco blocking hook.

Results. The headline result is computational: event-triggering reduces the number of online planner solves from 331 ± 3 (C1) to 29 ± 1 (C3) per 40 s run — an $11\times$ ($\approx 91\%$) reduction — at statistically indistinguishable tracking error (0.065 vs. 0.064 m). Per-solve cost is comparable (6.2 vs. 9.0 ms), so the saving is in solve *count*, not per-solve time (Fig. 2). Table II reports all metrics from fresh runs.

We are explicit about what the occlusion experiment does *not* show: when the EKF ArUco updates are suppressed, the covariance grows the same way for both configurations (peak $\mathrm{tr}\,\Sigma \approx 0.17$ in the injected window for C1 and C3 alike; Fig. 3), because the injected loss is independent of the UAV's replanning strategy. Both recover once the measurement returns. Thus the occlusion scenario demonstrates that the stress is real and the estimate stays bounded and recovers, but it does *not* evidence a covariance advantage for C3 — the demonstrated benefit of the event trigger is computational. The larger full-run peak $\mathrm{tr}\,\Sigma$ in Table II (0.245/0.197) is dominated by the initial EKF-convergence/takeoff transient rather than the occlusion, and it carries a large run-to-run standard deviation (± 0.16); we therefore report the occlusion-window peak, which isolates the injected-occlusion response and is tightly repeatable, as the covariance metric of record.

TABLE II

METRICS FROM FRESH SIMULATION RUNS (MEAN $\pm$ STD OVER R=4 REPEATS $\times$ {NOMINAL, OCCLUSION}= 8 RUNS/METHOD). AUTO-GENERATED INTO RESULTS/TABLE_1_SUMMARY.CSV (TABLE AND PROSE SHARE ONE WINDOW). †STEADY-STATE WINDOW $t \geq 8$ S (EXCLUDES THE TAKEOFF/EKF-CONVERGENCE TRANSIENT). ‡FULL-RUN PEAK IS DOMINATED BY THAT INITIAL TRANSIENT; THE OCCLUSION-WINDOW PEAK ISOLATES THE INJECTED-OCCLUSION RESPONSE AND CARRIES THE "BOTH REMAIN BOUNDED" POINT.

Metric	C1 (Continuous)	C3 (Event-Trig.)
Planner solves (#)	331±3	29±1
Mean tracking error (m)†	0.065±0.001	0.064±0.001
Mean $\mathrm{tr}\,\Sigma$†	0.0148±0.0084	0.0148±0.0084
Peak $\mathrm{tr}\,\Sigma$ (full run)‡	0.245±0.161	0.197±0.166
Peak $\mathrm{tr}\,\Sigma$ (occlusion window)	0.168±0.001	0.170±0.001
Mean UAV solve time (ms)	6.2±0.2	9.0±0.2

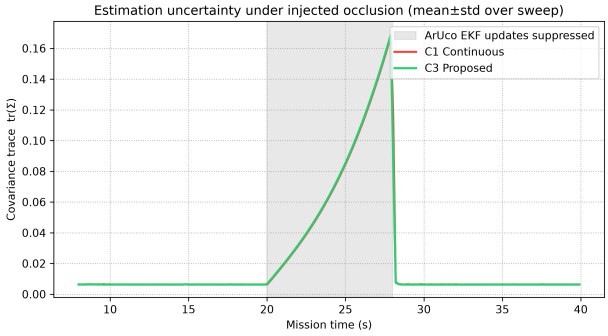


Fig. 3. Estimation uncertainty $\mathrm{tr}\,\Sigma(t)$ under injected occlusion (ArUco EKF updates suppressed over the shaded window), mean±std over R=4 repeats, after the takeoff transient. Both C1 (continuous) and C3 (event-triggered) exhibit the same covariance growth and recovery: the injected loss is independent of the replanning strategy. The event trigger's advantage is the $11\times$ reduction in solve count (Table II), not occlusion robustness.

VIII. CONCLUSIONS

We presented an observability-driven, event-triggered tube-MPC design for aerial–ground tracking and a full SITL integration in ROS 2/Gazebo. Our validated result is computational: the perception-driven trigger reduces online planner solves by $\approx 11\times$ at statistically equal tracking accuracy over 8 paired runs. We are deliberate that this is the *demonstrated* payoff — the trigger is observability-aware by design, but its isolated observability benefit is not established on the present platform, because the UAV tracks tightly enough that the FoV margin is not stressed and the tighter-FoV regime that would stress it is not trajectory-reproducible under the current non-lockstep simulator. Future work: a controllable-lag regime and a deterministic lockstep simulator to test the

observability claim; and, on the theory side, wiring the offline mRPI online, a homothetic tube with recursive-feasibility guarantees, a σ-polytopic fractional trigger, and hardware validation.

REFERENCES

[1] D. Q. Mayne, M. M. Seron, and S. V. Raković, "Robust model predictive control of constrained linear systems with bounded disturbances," *Automatica*, vol. 41, no. 2, pp. 219–224, 2005.

[2] S. V. Rakovic, E. C. Kerrigan, K. I. Kouramas, and D. Q. Mayne, "Invariant approximations of the minimal robust positively invariant set," *IEEE Trans. Autom. Control*, vol. 50, no. 3, pp. 406–410, 2005.

[3] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, "Constrained model predictive control: Stability and optimality," *Automatica*, vol. 36, no. 6, pp. 789–814, 2000.

[4] D. Falanga, P. Foehn, P. Lu, and D. Scaramuzza, "PAMPC: Perception-aware model predictive control for quadrotors," in *IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2018.

[5] R. Platt, R. Tedrake, L. Kaelbling, and T. Lozano-Pérez, "Belief space planning assuming maximum likelihood observations," in *Robotics: Science and Systems (RSS)*, 2010.

[6] J. van den Berg, S. Patil, and R. Alterovitz, "Motion planning under uncertainty using iterative local optimization in belief space," *Int. J. Robotics Research*, vol. 31, no. 11, pp. 1263–1278, 2012.

[7] A. Eqtami, D. V. Dimarogonas, and K. J. Kyriakopoulos, "Event-triggered control for discrete-time systems," in *American Control Conference (ACC)*, 2010, pp. 4719–4724.